SIEMENS

Opcenter™ Execution Core Release Notes: 2510

Rev. 1

Notices

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Document History
Release Notes: 2510
October 2025
Printed in the U.S.

Contents

Overview of this Release **1-1**

What's New in Release 2510 ..... **1-1**

Technology Enhancements 1-1

Functional Enhancements 1-1

Additional Enhancements 1-3

Where to Apply this Release **1-4**

Release 2510 (Metadata Version 10.5.0.0) **1-4**

Known Issues **1-5**

Known Mobile Device Issues 1-8

Problem Resolutions **1-9**

System Requirements **1-17**

License Management 1-17

Hardware Platform 1-17

Operating System 1-18

Database Management System (DBMS) 1-18

Web Servers 1-18

Mobile Devices 1-19

Client Environment 1-19

Third-Party 1-19

SAP Business Objects BI 4.3 SP 2 1-19

Microsoft 365 Access Run-Time Prerequisite 1-19

Before Installing the Latest Release **1-22**

Installing the Latest Release **1-23**

Configuring Industry Solutions Features and Modules **1-23**

Upgrading from Camstar Enterprise Platform 5.8, 6, or 7 ..... **1-23**

Updating Application Defaults for Secure by Default 1-23

Updating the Scale.js File for SignalR 1-24

Converting from Release 8 or Earlier **1-25**

When Upgrading from Release 8 or Version 7 1-25
Converting Your Metadata 1-25
Updating Your Database 1-26
Completing the Installation 1-27
When Upgrading from a Release of Version 6 1-27
Converting Your Metadata 1-27
Updating Your Database 1-28
Completing the Installation 1-29
When Upgrading from Release 5.8 1-29
Converting Your Metadata 1-29
Updating Your Database 1-30
Completing the Installation 1-31
When Upgrading from Earlier Versions of Camstar Enterprise Platform 1-31
Updating the Log In Page Bookmark if Upgrading from Release 8 through Release 2310 1-32
If Upgrading from Release 2310 or Later and Using the Mendix-based UIs with SignalR 1-32
User Documentation 1-33
Administration Guides 1-33
DataStore Guides 1-33
User Guides 1-34
Technical Reference 1-36
Opcenter Execution Core Intelligence 1-36
Scheduling 1-37
Industry Solutions 1-37
SPC 1-37
Manufacturing Operations Management 1-37
Additional Documentation 1-38
Siemens Support Center 1-39
Installation assistance 1-39
Security Information 1-39
Data Protection By Design of DI SW MOM Products and Solutions 1-39
Change History 1-41

Overview of this Release

Opcenter Execution Core (Opcenter EX CR) is an all-inclusive manufacturing execution system (MES) product that delivers the robust manufacturing functionality you expect and some optional shop floor and quality modules.

Release 2510 is a main release. It is considered an update release if you are using any previous version of Releases 2210 through 2504. Therefore, you can install 2510 as an upgrade to Releases 2210, 2304, 2310, 2404, 2410, or 2504 of Opcenter EX CR. If you are using any previous release of Versions 6, 7, or 8, you must uninstall and then install Release 2510 directly.

Important: All installed Opcenter EX CR products must be on the same release, including the same main and update release. Custom applications consuming the Camstar Server Connection Library (Camstar.Utility.dll) and/or the Camstar Insite XML Client Library (InsiteXMLClient.dll) must be using the same release of these modules as the server release they are transacting against.

What's New in Release 2510

Siemens Industry Software Inc. continues to improve Opcenter Execution Core with Release 2510.

Technology Enhancements

This release provides the following technology enhancements:

- Chrome 140+
- Edge Chromium 140+
- Dynamsoft 10.4.3100
- TinyMCE 6.8.4
- Microsoft Access Database Engine 2016 replaced with Microsoft 365 Access Runtime
- Updated Progress DataDirect ODBC wire protocol driver for Oracle and SQL Server
- Mendix 10.24.4

Functional Enhancements

This release provides the following new or improved capabilities:

- **Electronic Procedures: Mendix Module**
  - Redesigned and optimized operator terminal user interface.
  - Supports the most critical task types with electronic signatures such as:
    - Instruction tasks - pass/fail, acknowledgement, data collection.
    - Transaction tasks - move in, move, collect data, component issue.
  - Enables traditional process timers and a new task timer.
  - Introduces support for task-specific material lists for component issues.

- **Hold Management: Mendix Module**
  - Operator interface that allows initiating and viewing holds for the following items:
    - Container
    - Manufacturing order
    - Product
  - Supports optionally defining a custom hold message that is displayed to the operator when attempting to process a held item.
  - Provides the capability to search for items currently on hold and release one or multiple items in a single, streamlined transaction.
- **Artificial Intelligence**
  - **Modeling Copilot**
    - AI-powered copilot which greatly simplifies modeling configuration.
    - Converts PDF work instruction documents into a preview of the proposed modeling structure.
    - With a single click, you can automatically create the corresponding modeling data.
    - Prompt management enables customers to customize the AI prompts for different document types.

  Note: This is available only through an early adopter program.
- **Dynamic Purge Tool – GA**
  - Supports purging data from the transaction database.
  - Can optionally archive it into a designated schema.
  - Provides the ability to restore the archived data based on:
    - Unique instance identifiers
    - Batch execution identifiers
  - Supports automation through Microsoft SQL Server agent jobs to enable scheduled purging and archiving operations.
  - Available for Microsoft SQL Server only.

- **Observability**
  - Enables on-premise performance monitoring and observability through OpenTelemetry.
  - Allows the use of OpenTelemetry in place of Azure Application Insights for Azure monitoring.
- **Valid Substitutes Enhancement**
  - New option to simplify the assignment of any revision of a product as a valid substitute.
  - The product modeling instances have a new option to mark that any revision of that Product can be a valid substitute.
  - The material list modeling instance has a new option to allow all revisions as substitute.
  - If either option is enabled, then any revision of the product is a valid substitute.
- **Power BI Reports**
  - Delivers a new set of reports:
    - Component Where-Used
    - Component Traceability History
    - Lot Sampling History
    - Event Details
- **REST API Enhancements**
  - Extends action attributes to shop floor transactions.
  - Delivers an optimized Swagger schema.
  - Adds the ability to manage multiple data points in a single user defined data collection (UDCD).

Additional Enhancements

This release provides the following additional enhancements:

- Introduced enhancements to the Ansible playbooks.
- Supports a containerized deployment of the Mendix portal on AWS and Azure.
- Enables single sign-on with Mendix via security assertion markup language (SAML) based identity providers (IdP).
- Can navigate seamlessly from the Portal to Mendix.

Where to Apply this Release

Apply this release to the machines where the following Opcenter Execution Core components are installed:

- Application Server
- Management Studio
- Designer
- Portal Studio
- .NET Client
- DataStore
- Conversion Tool
- WCF Client
- Portal: Runtime (IIS) and Development (VS.Net)

Release 2510 (Metadata Version 10.5.0.0)

The following information is specific to Release 2510:

- Requires conversion if you are currently using Release 8 or earlier. See "Converting from Release 8 or Earlier" on page 1-25 for procedures.
- If you have Opcenter EX CR installed:
  - If a metadata conversion is required, you must regenerate WCF Services (after mdb conversion).
  - The Portal only requires a refresh of the pages after WCF Services are generated.
  - Refer to the table under "Problem Resolutions" on page 1-9 for a list of known problems recorded and resolved.

Known Issues

The following are known issues in this release:

- **Technology:**
  - Oracle: The application does not have millisecond precision for Oracle.
- **Installation:**
  - Antivirus software could interfere with your installation. You may have to exclude certain folders from antivirus scanning during installation.
- **Portal Studio 2.0:**
  - When updating XML via the XML mode, you must save your changes in XML mode and then save again in controls mode.
- **Logging into the Portal:**
  - Logging into the Portal on two separate tabs as two different people in the same browser session is not supported.
- **Modeling:**
  - Workflow modeling object – You cannot drag and drop a Spec or Workflow modeling object on the lower portion of the Workflow Diagram canvas when the vertical scroll bar is visible. You cannot scroll to the bottom of the canvas and drop the Spec or Workflow modeling object.

    The workaround, when the vertical scroll bars are visible and the zoom level is 100% or higher, is to drop the Spec or Workflow on the top portion of the workflow canvas. You can also zoom out to 50% in order to drop the Spec or Workflow onto the canvas.
  - The application displays an error when you try to save a modeling object group after removing a reference to a deleted instance. For example, when you delete a resource that is a member of a resource group, the application displays a reference to the deleted instance in the resource group. If you delete the reference to the deleted instance and click Save, the application displays an error. The workaround is to remove the modeling object instance from the modeling object group before deleting the instance.
  - When using the special character "<" in a modeling instance name, the application truncates the display of the modeling instance name so that only the text after the "<" sign appears. For example, if you enter Acme<1234 as the modeling instance name, the application truncates the modeling instance name to 1234 when displayed. The modeling instance name stored in the database is not truncated.
- **Shop Floor:**
  - Container names with spaces before or after – The application does not correctly account for spaces before or after a container name when you are running against an Oracle database. For example, you can start a container with the name CONT123. You can then start a container with the name CONT123_. In the second instance, the underscore indicates a space. The application allows you to save the second instance with the space at the end.

However, when you open another shop floor page, the application displays an error. The workaround is to avoid using spaces at the beginning or end of a container name.

- Container Search:
  - The application ignores underscores in a wildcard search when the underscore is the last character before the wildcard (for example, ABC1000_%). The application returns more results than expected.
  - When searching for a container that includes a square bracket or any special character in the container name, you may receive unexpected results if you are using SQL Server. Refer to the *Opcenter Execution Medical Device and Diagnostics Shop Floor User Guide* or the *Opcenter Execution Core Shop Floor User Guide* for information on a workaround.
- Electronic signatures – When an electronic signature requirement has been configured for a shop floor transaction at a spec, and the transaction is executed as a transaction task in an electronic procedure, the application prompts the user for a signature as expected.

  If there is also a training record with a status of “Allow with Signature” configured at the transaction task, the application displays an error message that not all electronic signatures were met. The application does not prompt the user for a signature for the training record because the task was already executed in the background.

  Siemens recommends customers review and verify transactions running background services or business rules to be sure they execute successfully.
- Start Two-Level page – The application does not increment container numbering correctly when:
  - You attempt to start parent and child containers,
  - Generate Names Automatically is set to True,
  - Generate Child Names Automatically is set to True, and
  - The Numbering Rule is configured with “Low Volume / No Skip.”

  The workaround is to start one container at a time or to configure the Numbering Rule with "High Volume / Skip."

- **Opcenter Execution Core Intelligence (Google Chrome and Microsoft Edge Chromium):**
  - The Business Objects login will fail when using "localhost" as the host name. Use the fully qualified domain name as the host name.
- **Inline Statistical Process Control (SPC):**
  - The SPC Common Component must be uninstalled manually when you uninstall Opcenter Execution Core.

    Please be advised that Siemens has identified other issues with Inline SPC. To obtain a resolution, please contact Product Support for a field fix.

- **Change Management package approval workflows:**
  - Camstar change management workflow - The application displays information in the Instructions for Approver field for the first approvers by default. The application does not update the Instructions for Approver information when you select a role and employee name for a different approver.
  - PLM change management workflow - Only the package owner can record the PLM approval decision. Do not assign a substitution role.
- **Document Attachment feature when used with single sign-on through SAMAuth and Teamcenter Share:**

  The Upload to Share functionality is not working in the release due to a known issue. The Upload to Share button will not be visible even after setting the; TeamcenterShareEnabled' parameter to True.

  This issue only affects the Document Attachment feature when combined with single sign-on through SAMAuth and the Teamcenter Share functionality. Otherwise, the Document Attachment page works as expected. This issue will be resolved in an upcoming release.
- **Database Encryption:**
  - There is a known issue with the Management Studio Encrypt Database Traffic feature for Oracle. If you select the Encrypt Database Traffic check box and specify Oracle as the Transaction Database Type or DataStore Database Type, the cwallet.sso file is ignored if you specify a path and the cwallet.sso file in the Certificate Store File field. Management Studio will look in the local Microsoft Certificate Store for the self-signed database certificate or the certificate authority trusted certificate that signed the database certificate that you are attempting to securely connect to.

  The workaround is to load the self-signed database certificate or the certificate authority trusted certificate that signed the database certificate into the local Microsoft Certificate Store.
- Converting to 2510:
  - REST API: When upgrading from Release 8.9 or earlier to Release 2210 or later, you must use the REST API tab in Management Studio to enter your credentials. This tab was introduced with Release 2210.
  - You must uninstall Closed Loop Manufacturing (CLM) before upgrading to the latest release. Install the new release of the application and upgrade CIO before installing the new version of CLM and continuing the configuration.

Known Mobile Device Issues

The following are known issues with mobile devices in this release:

- Online help is not available.
- All shop floor pages in Horizon theme work on mobile devices. Note that more complex pages, such as EProcedure, may be more difficult to use on mobile devices with smaller screens. Siemens does not guarantee a shop floor page will work on a mobile device if your organization has customized the page.
    - EProcedure will not work on an Android mobile device when the width of the screen is less than 1024 Density-independent Pixels (dp). The device should have at least 1024 dp of width.
- SPC: Rendering issues may occur when using SPC on mobile devices depending on screen size and resolution.
- Mobile bar code scanning: When mobile bar code scanning is on and the user sets focus on a date field with a date picker, the bar code scanning icon appears on top of the date picker icon.

  The workaround is to remove the focus from the date picker field. Users can click on another field or an empty area to remove the focus from the date picker field. The date picker icon becomes visible again. The user can then click on the icon to open the date picker.
- Siemens does not support the Export capability on mobile devices for this release.
- Single sign-on (SSO) is not supported in mobile mode.

Problem Resolutions

This section describes known problems recorded and resolved from the last release to this release. TAC or Salesforce numbers are provided if available.

This table describes the issues resolved in Release 2510.

ID	Description
TAC 9724657	On the Manage P/E tab, when a comment was updated, the timestamp for all previous comments was also being overwritten with the latest timestamp. **Resolution:** Updated the code so that the correct timestamp is displayed for each comment, and only the newly added or updated comment indicates the latest timestamp.
TAC 9865013	Unable to Translate Portal's Menu and MenuItem When Using Custom User Label. **Resolution:** Added a note to indicate that user labels cannot be used for the display of menu items to the *Opcenter Execution Medical Device and Diagnostics Portal Studio 2.0 User Guide* and the *Opcenter Execution Core Portal Studio 2.0 User Guide*.
TAC 9978833	In the Maintenance Management page, rows are not highlighted on selection and a warning is displayed asking users to select at least one row. **Resolution:** Updated the code to correctly highlight the row when selected.
SF 01357053	Expiration Date Offset on Container Maintenance Page when selecting different Timezone. **Resolution:** Updated the time zone on the Container Maintenance page to default to the current time zone.
SF 01382243	No NonScheduleTime has been calculated for this OEE Availability. **Resolution:** Corrected the SQL function that is calculating the OEE Availability to properly handle NonScheduleTime configured in the Mfg Calendar Shift.
SF 01399077	Unable to add the computation value with an expression result to the container attributes. **Resolution:** Updated the code, and now the computation values are added with an expression result to the container attributes.
SF 01415185	When using SSO, the session times out after 30 minutes even though the session is not idle. **Resolution:** After the recent SSO design changes, the maximum session timeout is now limited to 100 minutes. Customers can configure the session timeout up to this limit.
SF 01420420	Multiple clicks of the SSO button inadvertently created multiple user sessions and led to unnecessary consumption of available licenses. **Resolution:** Disabled the link for "Log in via SSO" after one use. Now, after one click, multiple user sessions are not created even if the user clicks the button multiple times. This link is enabled only when the page is refreshed or loaded again.

ID	Description
SF 01430964	Users are logged on with UTC instead of local time using SSO. **Resolution:** Updated the code to fix Timezone offset issue for SSO users.
SF 01447749	Error of "StringReplace - Value 〈/Message〉, Replace { with 〈" throw in Event log file **Resolution:** Updated the CIOCsutomProcPlugins.dll to only write errors to the event viewer when performing StringReplace function.
SF 01453812	Container Maintenance grid not cleared. **Resolution:** Updated the Container Maintenance page to clear the grid when a new search is executed.
SF 01460063	When attaching a document from the side bar, the pop-up closes but the success message stays. The message does not fade nor does the X in the corner close it. **Resolution:** Updated the attach document functionality to properly close the success message.
SF 01460176	Material Queue Fixes. **Resolution:** Updated the Material Queue validation logic to detect a Material Queue associated to an Operation, Mfg Order, or Container in addition to a Resource.
SF 01460867	Product Not Issued Error. **Resolution:** Updated the Material Queue consumption logic to check if all required materials have been issued and then bypass the Validate Material Consumption check.
SF 01466444	Component Replace transaction records the transaction history on the wrong container. **Resolution:** Updated the query used by the Mfg Audit Trail to properly retrieve the transactions against issued containers.
SF 01469335	SPC inline charts have been applying control limits based on 2 sigma rather than the correct 3 sigma standard used in Control Charts. **Resolution:** Updated the existing EvaluationType parameter to support a new value. That new value is “ASQ.” Added the AsqSigmaCount parameter to the SPC Chart Type modeling object. This parameter is the sigma multiplier when the EvaluationType parameter is set to a value of “ASQ.” In order for the control limits to be calculated automatically for all evaluation types, the parameter CapabilityStudyType must be set to a value of “ShortTerm.” You will also need to set the CalculateControlLimits parameter to “true.”
SF 01483570	Opcenter is not properly detecting and handling the revision string when it contains the “delimiter” character – which is configurable in Portal Studio. **Resolution:** Updated code to properly support revision values that contain the delimiter character.
SF 01488526	Unrecoverable "Invalid User or Password" Error After Database Connectivity Loss and Restoration. **Resolution:** Changed the code to detect any unexpected exception, free the associated database connection, and create a new connection automatically.

ID	Description
SF 01491139	About BOM Item Import: Field "Product" requires input. **Resolution:** Changed the required value to "Not Required" as Siemens now supports passing in the UnverifiedProductName field and there is CLF logic to verify that some value for a product is passed in.
SF 01496452	Change Management screen error. **Resolution:** Corrected the isUserDataCollectionDef_VP to prevent the error seen on save.
SF 01499934	Flexible CN EL Driver fails to restart after SFM shutdown. **Resolution:** Updated Shop Floor Resource Configuration (SRC) to return the Resource Name instead of the AssetNumber.
SF 01511011	Component Replace service does not validate product of container matches material item. **Resolution:** Update ComponentReplace validation CLF's to check that any supplied container is for a valid product of the material list item being replaced.
SF 01523821	Error during Move In in Production Client related to use of subworkflows. **Resolution:** Update SetContainerPrevStep CLF to check if there is a WorkflowStack available on the container and see if the last step is in the previous workflow.
SF 01525272	Cannot open a Modeling Resource from Where Used popup. **Resolution:** Update the code-behind to detect that there is no associated modeling page and to not include the reference in the grid.
SF 01527793	Status Icon shows wrong container status. **Resolution:** Update the portal page to use property conditions against the integer Status value instead of the text status value.
SF 01528222	When transaction logs are active -> value was either too large or too small for an Int32 for all pages. **Resolution:** Changed the property to a long (Int64).
SF 01533647	When the user enters a Qty value using a comma (,) decimal separator, the value is not properly handled and the number is treated as if there was no decimal point. **Resolution:** Updated the portal page and code-behind file to properly handle the entered value and pass the proper Qty value into the transaction.
SF 01535996	LMD Documentation PR - Read Committed Snapshot Isolation. **Resolution:** Updated important note in the *Opcenter Execution Medical Device and Diagnostics System Administration Guide* and the *Execution Core System Administration Guide* to indicate that you should consult with database experts before enabling Microsoft SQL Server Read Committed Snapshot Isolation (RCSI) on your database.

ID	Description
SF 01540177	Opcenter Change Management transactions display the History Mainline records without the HistoryId. The null value appears instead of the History Id value. **Resolution:** Updated the code to display the History Id for Change Management transactions.
SF 01540668	Error message about field length when opening Modeling screen for certain Process Timers. **Resolution:** Updated the Step, Workflow, and TaskList fields to use DisplayName as the field type.
SF 01541211	Description: Invalid XML Node: expected "Correct Time Zone", received "Unknown Time Zone: Europe/Istanbul." **Resolution:** Removed the leading space before the Europe/Instanbul timezone in the TZ Data file.
SF 01547532	Opcenter REST API ProductBOMMaint unable to update to Revision "1." **Resolution:** Introduced code changes that fixed the update logic by disabling the comparison of "Name" and "Revision" when a server model already exists. These fields are always sent—even if unchanged.
SF 01550572	Mfg Order changes Factory flag and results in error during order interface. **Resolution:** Updated the CLF logic to not modify a field on the Factory and then change it back to the original value.
SF 01554843	When updating an IIS binding, such as adding or removing a certificate, the Security Service application pool will not start successfully and returns “Security service: Database is down. Please try again later.” This occurs only when using Oracle as the backend database, and specifically when database encryption is enabled, “Validate Server Certificate” is enabled, but no certificate file is set in Management Studio. **Resolution:** The issue was caused by a null reference error when the CertificateFile property was unset in Management Studio. A null propagation check was added to the code referencing CertificateFile.
SF 01555905	Sticky Capable enabled text box fields on StartPage have bad visual appearance. **Resolution:** Updated CSS classes to properly apply the sticky styling to a "submit" type input and not apply the textbox class.
SF 01561567	Not all menu items are shown when opening portal with aspx url. **Resolution:** Updated to properly find child sub-menu for any depth of menu.
SF 01582532	Containers on Carriers don't get disassociated correctly when Qty is changed to 0 and the container is closed. **Resolution:** Updated the logic on the ChangeQty FinishModifyEntity event to properly clear the Carrier association from the child container that has been disassociated.
SF 01589540	Export metadata crash during comparison. **Resolution:** Fixed in the metadata export to handle situation where a CLF has duplicated entries.

ID	Description
SF 01595997	Multiple issues with Material Setup in Production and Assembly Client. **Resolution:** Corrected code-behind to always set the ResourceId and ResourceName if specified in the Serial Number Entry. Updated JavaScript code to detect an item has been assigned Pending Issue Details and find the next matching item with no pending issue details.
SF 01597288	Online Traveler shows wrong date format. Day and Month are switched if day < 10. **Resolution:** Specified the MoveIn and MoveOut header type as a String to avoid multiple conversions of the value to a "date" - which can end up transposing the month and day value.
SF 01600087	Failed to import isManufactuingCalendar.xlsx when the Team column is empty. **Resolution:** Removed unnecessary code that was resulting in the error being seen.
TAC 10270829	Increase scale of BOM quantity field. **Resolution:** Updated BOM modeling section in the *Opcenter Execution Medical Device and Diagnostics Modeling User Guide* and the *Opcenter Execution Core Modeling User Guide* to state that the Quantity Required Field is limited to four digits.
TAC 10377795	On the Message Center screen, the Clear All button clears all the fields, including the Organization field, which is not supposed to be cleared. **Resolution:** Updated the code and now the Clear All button clears the values from all fields on the Message Center page, except for the Organization.
TAC 10510571	Need to Configure Time till Device Connection Stop. **Resolution:** Updated the JavaScript for the Material Request Status Board page to provide a better connection disconnect detection and automatic re-connect.
TAC 10877057	Test emails cannot be sent due to an EmailQueue table error and authentication failure with the email server. **Resolution:** Updated the code to validate user credentials for authentication when the SMTP mail server is configured for Email Notification Service.
TAC 11069783	Login Errors Caused by Expired Oracle Sessions. **Resolution:** Updated the code to detect the ODBC failure and automatically establish a new database connection when an Oracle session has expired.
TAC 11102364	Availability Calculation for OEE. **Resolution:** Updated the *Opcenter Execution Medical Device and Diagnostics Industry Solutions User Guide* and the *Opcenter Execution Core Industry Solutions User Guide* to point to the correct table that calculates the availability of a resource.

ID	Description
TAC 11131010	Issues when total number of resource to modify is more than the Max Rows Returned setting has allowed. **Resolution:** Enhanced Maintenance Activation page by allowing the activation or deactivation of each maintenance class a maximum of 5000 records per submission even though the number of resources under that maintenance class exceeds the maximum number. User can refer to the newly added Activated count column in the Maintenance Classes grid to check the number of resources under that maintenance class that has been activated and Resource Count column to see the total number of resources configured to the maintenance class.
TAC 11137748	The current system has a security vulnerability due to the ability to input HTML tags and JavaScript in critical fields, such as Container Name and Modeling Name. This vulnerability allows for Cross-Site Scripting (XSS) attacks, where malicious scripts like <script>alert('xss')</script> could be executed in other users' browsers. **Resolution:** • Added XSSPreventionEnabled property in Portal Studio Default Settings (default: True). • Implemented pattern-based detection whch blocks dangerous HTML tags, event handlers, and JavaScript protocols. • Preserved commonly used special characters that users need for workflows. • Show a clear warning: "Please enter plain text only. HTML tags and scripts are not allowed." • Users can disable it at their own risk for backward compatibility.
TAC 11140726	Incorrect Expiration Date format in Training Record Management page. **Resolution:** Cleared out the dateFormat setting in relatedin PackageSearchMultiple_VP, TrainingRecordManagement_VP, OrderDispatchVP, MfgCalendarMaint_VP and PackageInquiry_VP. The Dateformat now is an Out-Of-Box (OOB) setting.
TAC 11215082	Http 500 after rebuilding REST API - REST API services did not start correctly. **Resolution:** Updated note related to users that are using a different domain than the default domain to the *Opcenter Execution Medical Device and Diagnostics System Administration Guide* and the *Opcenter Execution Core System Administration Guide*.
TAC 11215772	LMD: Cannot connect to license server. **Resolution:** Updated the “Using Opcenter EX MDD Release 2210 or Later through a Firewall” section of the *Opcenter Execution Medical Device and Diagnostics Security Guide* and the *Opcenter Execution Core Security Guide* with Siemens License Vendor information.
TAC 11217463	Configuring License Information in Management Studio Before Initial DBUpdate Will Cause System Error. **Resolution:** Updated Management Studio to detect whether the database has not yet been created when first saving the license information and alerting the user that they must update the database.

ID	Description
TAC 11243565	Expiration Date Offset on Container Maintenance Page when selecting different Timezone. **Resolution:** Updated the Container Maintenance Page to use the user's time zone for the Expiration Date.
TAC 11253466	ContainerMaint - Resource ProductionStatus is not cleared. **Resolution:** Added logic on the ContainerMaint StartMOdifyingEntity event to check if Resource is being changed— and if original Resource is InProcess to call events to clear line clearance for the container.
TAC 11253794	Change Qty shows invalid values in the Portal when hitting the Reset button. **Resolution:** Updated the code so that the Change Qty field shows correct values when you click the Reset button.
TAC 11258904	Need AKS 2304 images and AKS 2310 images with Win 2022 OS. **Resolution:** Created 2304.0002 and 2310.0002 images using Windows Server 2022 as the base image. App server with remote client gateway support is also added.
TAC 11259882	Error of "StringReplace - Value 〈/Message〉, Replace { with 〈" throw in Event log file. **Resolution:** Updated the CIOCsutomProcPlugins.dll to only write errors to the event viewer when performing StringReplace function.
TAC 11260488	Performance issues were noticed during Container Pod startup due to the use of Microsoft IISRESET. **Resolution:** Removed IISRESET and added support to use net start/stop command for starting and stopping IIS related services. This has been incorporated in the Opcenter-aspnet base image, to be used in the app server, security services, and Portal images.
TAC 11295424	After loading the Chinese dictionary, some items are still shown in English. **Resolution:** Provided updated Dictionary files that contain the appropriate LabelID values.
TAC 11297639	Document Modeling page is crashing. **Resolution:** Updated page to accept null BrowseMode.
TAC 11299965	Scrolling issue in Container search. **Resolution:** Updated the Container Search page to reset the scroll to top of the grid when a new search is executed.
TAC 11300569	Opcenter Core Transaction Tester Documentation Change Requested. **Resolution:** Updated licensing information in the *Opcenter Execution Core Transaction Tester User Guide*.
TAC 11304853	2504 installer does not include the required version of the C++ run-time redistributable. **Resolution:** Added Microsoft Visual C/C++ runtime version14.40.33810 (branded as "Visual Studio 2015—2022") to the installer.

ID	Description
TAC 11308327	Required field label text not changing color in object modeling screen. **Resolution:** Update web control code to properly remove the CSS class that sets the label text to a "disabled" color.
TAC 11311465	Online Traveler - String was not recognized as a valid DateTime. **Resolution:** Added new GetDateTime method to safely get DateTime value handling different formats.
TAC 11311769	RouteStep CDO (InstanceId="001b7f8000000002") not found. **Resolution:** Updated the ReplaceWorkspace application to properly handle the applying of fields to sub-entity lists from a lower workspace when a new column must be added.
TAC 11342984	Opening square bracket creates erratic behavior in Container Search when SQL Server is used. **Resolution:** Updated the *Opcenter Execution Medical Device and Diagnostics Shop Floor User Guide* and the *Opcenter Execution Core Shop Floor User Guide* with information on how to search for containers with special characters in container name.
TAC 11356826	When the National Character Set for the database is UTF8, while upgrading from V8 to 2504, the data in the table QueryText is getting corrupted. **Resolution:** The National character set on the Oracle database should be updated to AL16UTF16. The *Opcenter Medical Device and Diagnostics System Administration Guide* and the *Opcenter Execution Core System Administration Guide* have been modified to include the correct National Character Set for 2510.

System Requirements

This section details the minimum requirements and third-party prerequisites to install Opcenter Execution Core.

License Management

Important: Release 2510 leverages Siemens advanced licensing technology (SALT). The Siemens License Server (SLS) replaces the FlexNet license manager. The Siemens License Server was introduced with Release 2310. It supports new SALT-based license files and legacy FlexNet-based license files. You must install the Siemens License Server if you have not installed it previously.

You must request a new license file when upgrading to Release 2510 from a release prior to Version 7 of Camstar Enterprise Platform. You must contact your Customer Account Analyst (CAA) for the new license file before you install Release 2510. The license file is generated and sent to you by e-mail.

Note: Your current serial number is valid for Release 2510 if you are licensing the same products and modules. You must enter this serial number when installing the new version.

Refer to the *Opcenter Execution Core Installation Guide* for information on obtaining a new license file and installing the Siemens License Server.

Hardware Platform

These are the minimum hardware requirements for a Release 2510 installation. vCPU refers to virtual CPU (central processing unit):

Application Server	• 8 vCPU, 16 GB RAM • RAID 10 disk configuration recommended
Client	• 2 vCPU, 4 GB RAM
Opcenter Execution Core Intelligence Server (Business Objects)	• 8 vCPU, 16 GB RAM
Single user development environments require at least 4 vCPU and 8 GB RAM (more powerful machines result in better performance).	
Multi-user development environments require at least 8 vCPU and 16 GB RAM (more powerful machines result in better performance).	
Refer to the Release Matrix on the Support Center website for information about virtualization software.	

Operating System

Your Operating System (OS) must meet the minimum system requirements and must be installed in compliance with the manufacturer's design and specification.

- Windows Server 2019 Standard and Datacenter or Windows Server 2022 Standard and Datacenter. Desktop mode is required for a typical installation.

Important: Always refer to the product Release Matrix on the Siemens Support Center website for the latest information.

Database Management System (DBMS)

- You must have an operational Database Management System (DBMS) client installed on the server. The DBMS engine is not required on every server.
- Your database must meet the minimum system requirements and must be installed in compliance with the manufacturer's design and specification.
- You must define a user who can execute database transactions and connections. This user has no effect on, and does not depend on, any account defined within Opcenter Execution Core.

The required DBMS is one of the following SQL Server or Oracle options:

- Microsoft SQL Server 2019 or 2022 Standard and Enterprise

  These releases support an Always On Failover Cluster Instance (FCI) configuration which is an Active/Passive 2-node configuration for Failover for the SQL Server FCI that contains the OLTP database schema.

- Oracle 19c (12.2.0.3)

  This release supports non-container instances only. This release supports an Oracle RAC Active/Active configuration for failover for the RAC database that contains the OLTP database schema.

Important: You must install a SQL command line interface for SQL Server or Oracle. You must install sqlcmd for SQL environments. You must install SQL*Plus for Oracle environments. The batch scripts use sqlcmd or SQL*Plus to connect to the RDBMS to execute SQL scripts.

Note: Oracle RAC or SQL Server FCI configurations for DataStore databases are not currently supported.

Web Servers

Internet Information Services (IIS) 10 must be installed on all Web servers.

Mobile Devices

These are the minimum requirements for supported mobile devices:

- Android - Samsung Galaxy S23 or Galaxy Tab S9:
  - Memory - 16 GB RAM
- iOS- iPhone 15 or 11th generation iPad:
  - Memory - 12 GB RAM

Client Environment

- Google Chrome 140+
- Microsoft .NET Framework 4.8
- Microsoft .NET Framework 8.0.5 (for REST API)
- Microsoft Edge Chromium 140+
- Windows 11 Professional

Important: Although different browsers and versions of those browsers function similarly, differences may be experienced in response times and rendering of application pages.

Third-Party

SAP Business Objects BI 4.3 SP 2

This is required only for Operational Reporting in Opcenter Execution Core Intelligence.

Microsoft 365 Access Run-Time Prerequisite

Microsoft Access Database Engine 2016 Redistributable—used by the Management Studio component of Opcenter Execution Core and Opcenter Medical Device and Diagnostics —reached end of life on October 14, 2025.

You must build a deployment package for Microsoft 365 Access Run-Time using the Office Deployment Tool and a custom configuration file. You can then deploy the package to servers by uninstalling the legacy component, transferring deployment files, and then installing the new run-time.

Building the Deployment Package

To create the deployment package for Microsoft 365 Access Run-Time, follow these steps on an Internet-connected machine:

1. Download the Office Deployment Tool (ODT):

   Download ODT

2. Install ODT:

   Run the downloaded executable and extract it to: C:\ODT

3. Create a configuration file:

   In C:\ODT, create a new file named AccessRuntimeConfig.xml.

4. Insert Configuration XML:

   Paste the following content into AccessRuntimeConfig.xml:

```
<Configuration ID="07994891-3795-404e-9dd1-13cabc0e61cf">
    <Add OfficeClientEdition="64" Channel="Current"
    SourcePath="c:\AccessRuntimeDownload" AllowCdnFallback="FALSE">
        <Product ID="AccessRuntimeRetail">
            <Language ID="en-us"/>
        </Product>
    </Add>
    <Updates Enabled="FALSE" UpdatePath="c:\AccessRuntimeDownload"/>
    <Display Level="Full" AcceptEULA="TRUE"/>
</Configuration>
```

   Note: Ensure that SourcePath and UpdatePath point to a folder that exists on both the build and target machines. In this example, use C:\AccessRuntimeDownload.

5. Download installation files:

   Open Command Prompt in C:\ODT and run:

   setup.exe /download AccessRuntimeConfig.xml

   This will download the installer files to C:\AccessRuntimeDownload:

   - Download size: ~3.5–4 GB
   - Installed size on server: ~50 MB

Deploying the Package

To install Microsoft 365 Access Run-Time on each application server, follow these steps:

1. Close Applications:

   Ensure Management Studio and/or Designer are closed.

2. Uninstall the legacy component:

   Navigate to Control Panel > Programs and Features and uninstall Microsoft Access Database Engine 2016.

3. Transfer deployment files:

   Copy the following folders from the source machine to each application server:

   - C:\ODT
   - C:\AccessRuntimeDownload

4. Verify Folder Paths:

   Confirm that the folder paths are identical on each server.

5. Install Access Run-Time:

   On each server, open Command Prompt in C:\ODT and run:

   setup.exe /configure AccessRuntimeConfig.xml

6. Complete Installation:

   Once installation is complete, reopen Management Studio and/or Designer as needed.

Before Installing the Latest Release

Perform these tasks before installing the latest release:

- Check for available Windows Updates and install if available.
- Ensure that all Rivest Cipher 4 (RC4) ciphers are disabled in or removed from the registry on all clients and servers that will run Opcenter EX CR if you use TLS/SSL in your environment. RC4 ciphers can cause security vulnerabilities. Refer to this Microsoft security advisory for more information: https://support.microsoft.com/en-us/help/2868725/microsoft-security-advisory-update-for-disabling-rc4.
- When upgrading to Release 2510 from Release 8.3 or prior, if you use an Oracle database and have a DataStore database configured and you want to retain log messages in the DataStoreLog table, ensure that you save the log messages in the DataStoreLog table before upgrading to Release 2510 and performing a DBUpdate in Management Studio. The DBUpdate performed after the upgrade to Release 2510 truncates the DataStoreLog table before the data type of the Job column in the DataStoreLog table is changed from a NUMBER to a VARCHAR(128) data type.
- Uninstall Closed Loop Manufacturing (CLM). Install the new version of the Opcenter EX CR base system and upgrade CIO before installing the new version of CLM and continuing the configuration.
- Siemens highly recommends that you back up important files before proceeding.
- Depending on the version of the application from which you are upgrading, there are important considerations to take into account. Refer to this table to determine the sections that apply to your release.

If you are upgrading from . . .	Then refer to . . .
Version 5.8, 6, or 7	"Upgrading from Camstar Enterprise Platform 5.8, 6, or 7"
Release 8 or earlier	"Converting from Release 8 or Earlier"
Release 8 through Release 2310	"Updating the Log In Page Bookmark if Upgrading from Release 8 through Release 2310"
Release 2310 or later	"If Upgrading from Release 2310 or Later and Using the Mendix-based UIs with SignalR"

Note: More than one section may apply depending on the release you are using.

Important: Before proceeding with the installation process, use the Windows Control Panel to uninstall the Opcenter UMC Agent if you are upgrading from Release 2310 or earlier.

Installing the Latest Release

You must always install the latest release to ensure you have the latest features and functions.

Follow these steps to install a release:

1. Download the Release executable from the Siemens Support Center support website at:

   Opcenter Execution Core Downloads

2. Open the folder or zip file from Support.
3. Double-click the executable file and follow the directions from the installer.
4. Acknowledge required post-installation actions prompted by installer.

Configuring Industry Solutions Features and Modules

Industry Solutions features, such as Opcenter Execution Core Scheduling and Opcenter Execution Core Batch Processing, require configuration before you can use them. Information on running the Industry Solutions configuration tool is available in the *Opcenter Execution Core System Administration Guide*.

Upgrading from Camstar Enterprise Platform 5.8, 6, or 7

Siemens highly recommends that you set up a new server with Microsoft Windows Server 2019 or Windows Server 2022 installed and that you back up important files. You will also need your license file and serial number. Refer to the "Converting from Release 8 or Earlier" on page 1-25 for information on converting the metadata.

Consider the following changes you may need to make when upgrading from a Release 5.8, Version 6, or Version 7 installation of Camstar Enterprise Platform.

- Default security browser settings
- Update the scale.js file for SignalR

Your bookmarked URL for Release 5.8, Version 6, and Version 7 of Camstar Enterprise Platform continues to take you to the application's Log In page.

- https://servername/CamstarPortal/ directs you to the default.aspx page and displays the application in the Horizon theme.

Updating Application Defaults for Secure by Default

These are changes you will need to make unless you are disabling secure by default. Refer to the *Opcenter Execution Core Security Guide* for information on disabling SSL.

- Default browser security settings prevent documents on unsecured websites from opening. You must change your settings to allow unsecured content. Attempting to open unsecured content (http) within the Portal results in an error in the address bar or at the bottom of the screen depending on the browser.
- You must update client applications previously making TCP requests to the Application Server on port 2881 to make HTTP/HTTPS requests on ports 80/443.

- Change management:
  - Make sure the entries in the Server Catalog System for each server uses the correct DNS name in the IP Address setting. It needs to match the subject of the certificate (for example, AppServer1.appserver.local).
  - Ensure there is a valid SSL certificate (not localhost) on the target and source servers.
- SMTP transports:
  - The IP Address field on the SMTP Transport pages should be populated with the host name found on the SSL certificate instead of the IP address.
  - Ensure there is a valid SSL certificate (not localhost) on the SMTP transports.

Updating the Scale.js File for SignalR

You will need to make this change to the scale.js file if you have custom pages using SignalR.

Old Javascript Value	New Javascript Value
$.connection.cIOEventHub	$.connection.cIOEventSignalRHub

Converting from Release 8 or Earlier

Important: This section applies only to users who are currently using Release 8 or earlier and are converting to Release 2510.

Siemens highly recommends that you set up a new server with Microsoft Windows Server 2019 or Windows Server 2022 installed and that you back up important files. You will also need your license file and serial number.

You will need to convert the metadata following the installation of a release. Converting to Opcenter Execution Core Release 2510 is a multi-stage process. The process for converting the metadata depends on the previous version from which you are upgrading. Refer to the appropriate section for instructions:

- When Upgrading from Release 8 or Version 7
- When Upgrading from a Release of Version 6
- When Upgrading from Release 5.8

Refer to the chapter on converting to Release 2510 in the *Opcenter Execution Core Installation Guide* for more information.

Important: Make backup copies of important files before proceeding.

When Upgrading from Release 8 or Version 7

This section describes converting your metadata when you are upgrading from an earlier release of Versions 7 or 8.

Important: Additional steps are required to convert metadata if you have Medical Device workspaces installed. Refer to the *Opcenter Execution Core Installation Guide* for information.

Converting Your Metadata

Follow these steps to convert your metadata:

1. Back up existing InSite and SiteInfo mdb files from the previous version.
2. Install all necessary Opcenter Execution Core products and their components, and configure in Management Studio.
3. Copy the current and base metadata InSite.mdb from their respective folders into a temporary folder.
4. Rename these files so they can be used as input for the Replace Workspace Data Utility.

   Important: Do not update the InSite.mdb file in the C:\Program Files (x86)\Camstar\InSite Administration\Base Metadata folder. The application requires this file for the metadata upgrade process. Keep this file unmodified in the Base Metadata folder to ensure update installers can successfully replace the file with each update.

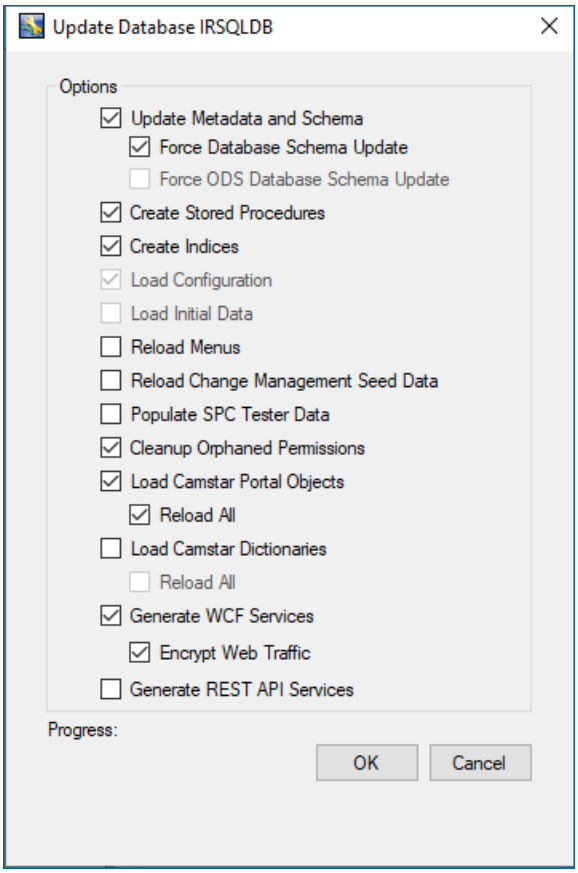
5. Run the Replace Workspace Data Utility.

6. Rename the upgraded InSite.mdb you created with the Replace Workspace Data Utility to the standard name and copy it to the default location.
7. Update the transaction database using “Updating Your Database.”
8. Complete the installation tasks using “Completing the Installation.”

Updating Your Database

Follow these steps to update your database:

1. Open Management Studio.
2. Expand the Databases node.
3. Right-click the Transaction Database and select **Update Database**. *The Update Database window appears.*
4. Select the check boxes shown in this image.



Important: Ensure the Update Metadata and Schema, Force Database Schema Update, Load Camstar Portal Objects, Reload All, and Generate WCF Services options are selected.

5. Click **OK**.
6. Complete the remaining installation tasks in the next procedure.

Completing the Installation

Follow these steps to complete the installation:

1. Open an administrative command prompt and navigate to %installdir%\InSite Administration\scripts.
2. Run csiPopulateCMUpdate.bat from the command prompt.
3. Run csiPolulateCMMenu.bat from the command prompt.

   Note: Database clients are required for script execution. You must install a SQL command line interface for SQL Server or Oracle. You must install sqlcmd for SQL environments. You must install SQL*Plus for Oracle environments.

   Refer to the Microsoft website for information on the system requirements for installing the sqlcmd utility.

4. Perform iisreset.
5. Clear the browser cache.
6. Generate WCF services.
7. Review permissions for Portal pages.

When Upgrading from a Release of Version 6

This section describes converting your metadata when you are upgrading from a release of Version 6.

Important: Additional steps are required to convert metadata if you have Medical Device workspaces installed. Refer to the *Opcenter Execution Core Installation Guide* for information.

Converting Your Metadata

Follow these steps to convert your metadata:

1. Back up existing InSite and SiteInfo mdb files from the previous version.
2. Install all necessary Opcenter Execution Core products and their components, and configure in Management Studio.
3. Copy the current and base metadata InSite.mdb from their respective folders into a temporary folder.
4. Rename these files so they can be used as input for the Replace Workspace Data Utility.

   Important: Do not update the InSite.mdb file in the C:\Program Files (x86)\Camstar\InSite Administration\Base Metadata folder. The application requires this file for the metadata upgrade process. Keep this file unmodified in the Base Metadata folder to ensure update installers can successfully replace the file with each update.

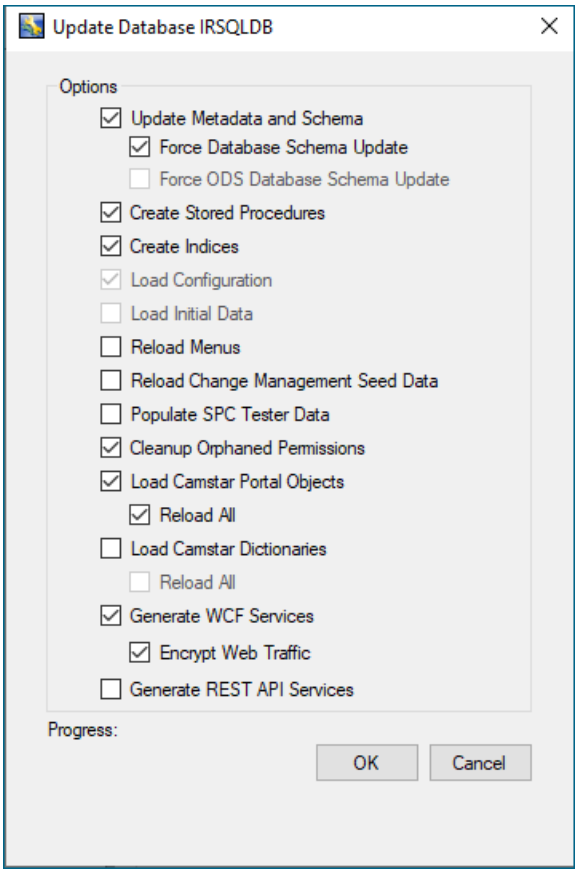
5. Run the Replace Workspace Data Utility.

6. Rename the upgraded InSite.mdb you created with the Replace Workspace Data Utility to the standard name and copy it to the default location.
7. Update the transaction database using “Updating Your Database.”
8. Complete the installation tasks using “Completing the Installation.”

Updating Your Database

Follow these steps to update your database:

1. Open Management Studio.
2. Expand the Databases node.
3. Right-click the Transaction Database and select **Update Database**. *The Update Database window appears.*
4. Select the check boxes shown in this image.



Important: Ensure the Update Metadata and Schema, Force Database Schema Update, Load Camstar Portal Objects, Reload All, and Generate WCF Services options are selected.

5. Click **OK**.
6. Complete the remaining installation tasks in the next procedure.

Completing the Installation

Follow these steps to complete the installation:

1. Open an administrative command prompt and navigate to %installdir%\InSite Administration\scripts.
2. Run csiPopulateCMUpdate.bat from the command prompt.
3. Run csiPolulateCMMenu.bat from the command prompt.

   Note: Database clients are required for script execution. You must install a SQL command line interface for SQL Server or Oracle. You must install sqlcmd for SQL environments. You must install SQL*Plus for Oracle environments.

   Refer to the Microsoft website for information on the system requirements for installing the sqlcmd utility.

4. Perform iisreset.
5. Clear the browser cache.
6. Generate WCF services.
7. Review permissions for Portal pages.

When Upgrading from Release 5.8

This section describes converting your metadata when you are upgrading from Release 5.8.

Important: Additional steps are required to convert metadata if you have Medical Device workspaces installed. Refer to the *Opcenter Execution Core Installation Guide* for information.

Converting Your Metadata

Follow these steps to convert your metadata:

1. Back up existing InSite and SiteInfo mdb files from the previous version.
2. Install all necessary Opcenter Execution Core products and their components, and configure in Management Studio.
3. Send your 5.8 InSite.mdb file to Product Support for conversion if you are upgrading from Release 5.8.

   Important: Do not update the InSite.mdb file in the C:\Program Files (x86)\Camstar\InSite Administration\Base Metadata folder. The application requires this file for the metadata upgrade process. Keep this file unmodified in the Base Metadata folder to ensure minor release installers can successfully replace the file with each minor release.

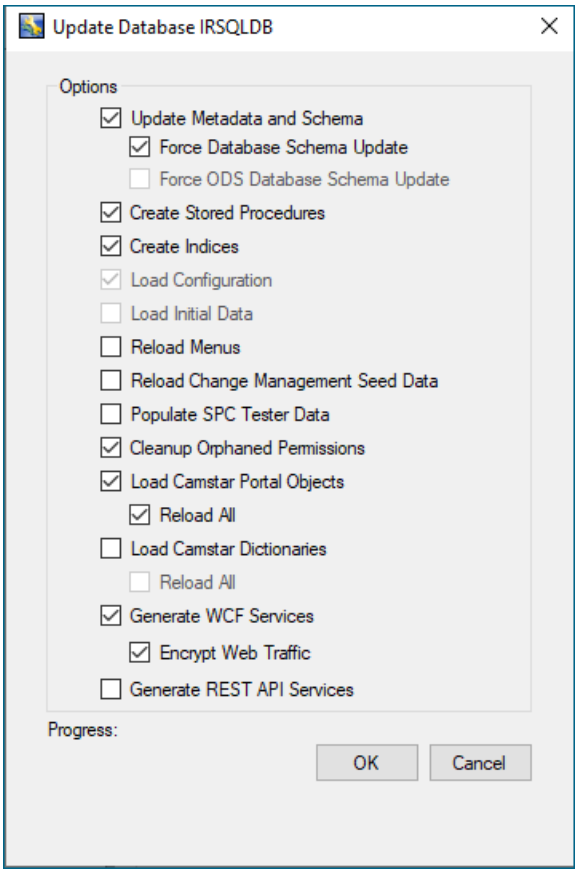
4. Update your test database with the converted InSite.mdb file Product Support sends you.
5. Regenerate the WCF Services.

6. Update the transaction database using "Updating Your Database ."
7. Complete the installation tasks using "Completing the Installation."

Updating Your Database

Follow these steps to update your database:

1. Open Management Studio.
2. Expand the Databases node.
3. Right-click the Transaction Database and select **Update Database**. *The Update Database window appears.*
4. Select the check boxes shown in this image.



Important: Ensure the Update Metadata and Schema, Force Database Schema Update, Load Camstar Portal Objects, Reload All, and Generate WCF Services options are selected.

5. Click **OK**.
6. Complete the remaining installation tasks in the next procedure.

Completing the Installation

Follow these steps to complete the installation:

1. Open an administrative command prompt and navigate to %installdir%\InSite Administration\scripts.
2. Run csiPopulateCMUpdate.bat from the command prompt.
3. Run csiPolulateCMMenu.bat from the command prompt.

   Note: Database clients are required for script execution. You must install a SQL command line interface for SQL Server or Oracle. You must install sqlcmd for SQL environments. You must install SQL*Plus for Oracle environments.

   Refer to the Microsoft website for information on the system requirements for installing the sqlcmd utility.

4. Perform iisreset.
5. Clear the browser cache.
6. Generate WCF services.
7. Review permissions for Portal pages.

When Upgrading from Earlier Versions of Camstar Enterprise Platform

Contact Product Support via the Support Center for conversion assistance.

Updating the Log In Page Bookmark if Upgrading from Release 8 through Release 2310

Important: Updating your Log In page bookmark is an important consideration when upgrading from Release 8 through Release 2310.

Opcenter Execution Core provides two URLs for logging in to the application:

- Horizon theme, the desktop computer default: CamstarPortal/default.aspx

  The Horizon theme has a user interface with left side navigation. It is intended to be viewed from a computer—not a mobile device. Shop floor pages render responsively by default.

- Mobile: CamstarPortal/default.aspx?mode=mobile

  The application displays this theme by default if you are accessing the application from a mobile device such as a phone or a tablet. It uses the Horizon theme with top navigation. Shop floor pages render responsively and resize appropriately for the device you are using.

Note: The default.htm URL used to log in to the Portal for Releases 8 through 2310 will be deprecated in an upcoming release.

If Upgrading from Release 2310 or Later and Using the Mendix-based UIs with SignalR

If you are upgrading from Release 2310 or later and you are using the Mendix-based UIs with SignalR, you must remove these entries in the web.config file in the Camstar Portal folder:

```
<!-- Already allow all origins in Startup.cs file.
    <add name="Access-Control-Allow-Origin" value="https://localhost:4040" />
    -->
```

Making this update enables you to leverage the SignalR Hub and connect to it from outside of the Portal.

User Documentation

The Opcenter Execution Core documentation library is available in PDF format on the Siemens Support Center website. Visit the Support site for the most current documentation: Technical Documentation for Opcenter Execution Core.

Acrobat Reader is required. You can download Acrobat Reader for free from the Adobe Systems website (http://www.adobe.com).

The following files are available for Opcenter Execution Core:

Document Name	Description
Release Notes: 2510	This document.
Detailed Release Notes: 2510	Provides more information about the release, specifically information about the metadata delta report, and the files added or changed.
Installation Guide **Updated!**	Provides the explanations, instructions, and illustrations you need to install Opcenter Execution Core products and modules. This includes procedures for conversion from earlier Camstar Enterprise Platform versions, pre-installation setup, installation procedures, and server setup.

Administration Guides

Document Name	Description
Security Guide **Updated!**	Provides information about security measures included in Opcenter Execution Core.
System Administration Guide **Updated!**	Provides information about using Management Studio, configuring the security model, and configuring e-mail notification. Also includes information on best practices, clustering, transaction tracing, and database identifiers.

DataStore Guides

Document Name	Description
DataStore Reference	Contains procedures to install and use the DataStore module, including summary tables, for database reporting purposes.

User Guides

Document Name	Description
Change Management User Guide	Contains information about the application's manufacturing process change management application and describes how an organization can export modeling objects from one system and then import those modeling objects on another system.
Data Collection User Guide	Procedures to create parametric data definitions using the Designer Data Collection Wizard. Also includes procedures for creating web parts for data collection and adding those web parts to the pages to use for data collection.
Designer User Guide	Information about the new and improved Designer, including configurable data objects (CDOs) and how they are used in object designing.
Event Management User Guide **Updated!**	Provides an introduction to the Event Management features in the Portal that allow you to record an event for a process and a failure, investigate and identify the source of the deviation, and determine and enforce final actions required to close the event.

Document Name	Description
Modeling User Guide **Updated!**	Contains information about defining the factory model using the Portal. Additional modeling pages are included in this guide for the optional modules: • AQL Sampling - Provides the ability to model and execute sampling plans for containers. AQL Sampling can be used to determine if a container is accepted or rejected. • Dispatch Management - Provides the ability to configure and display a list of work and the sequence in which the work must be performed to the shop floor user. You can enforce the rules on what types of lists are displayed by defining dispatch rules in the factory model. • Dispatch lists are used by shop floor users as the basis for starting or moving containers. • Electronic Signatures - Provides the ability to obtain approval electronic signatures for various shop floor transactions. An electronic signature assigns accountability for the performance of a distinct activity to the signer and is indicated as such in the associated audit trail of activity. • Label Printing - Provides the ability to print and reprint container and nonconformance labels, handle printing errors, and view summary and detail information for printed labels. • Maintenance Management - Provides the ability to define, enforce, and track Resource-centric maintenance requirements based on calendar date and thruput. • Operator Training and Certification - Provides the ability to establish manufacturing roles, defines training for the various roles, maintains training records, and establishes certification expiration within the system. Performing certification ensures that only qualified employees perform prescribed shop floor tasks. • Medical Device Workspace – Provides information on the features and modules available in the medical device workspace. Provides the ability to define and assign unique device identifiers (UDI) and the ability to record multiple users for a shop floor transaction. • Batch Processing – Provides the ability to control and track production of small chemical batches used as components of devices.
Nonconformance Management User Guide **Updated!**	Provides an introduction to the Nonconformance Management features in the Portal that allow you to detect and capture the initial observed failure and its nonconformance characteristics, investigate and identify the source of the deviation, and determine and enforce final actions required to close the nonconformance.

Document Name	Description
Portal Studio 2.0 User Guide **Updated!**	Contains information about the new and improved Portal Studio 2.0. Portal Studio 2.0 is a self-contained development and configuration tool used to customize and maintain the Portal.
Shop Floor User Guide **Updated!**	Contains information about performing Shop Floor transactions using pages delivered with the Portal. This guide provides information about the pages used to perform these transactions (also known as services) and explains how to use them to execute transactions that record and track information about containers, establish and maintain resources, monitor and control quality, and generate manufacturing audit trails. Additional transaction pages are included in this guide for the optional modules.

Technical Reference

Document Name	Description
Technical Reference: .NET APIs	Information about XML APIs; provides code samples for ActiveX. For use with the specific API specification.
Technical Reference: High Performance Engine **Updated!**	Describes how you can use the High Performance Engine (HPE) for performing business logic on the database rather than on the application server. Using HPE enables a large number of CDOs to be inserted and updated without excessive network chatter between the application server and database.
Technical Reference: REST API **Updated!**	Describes the REST APIs provided with the Siemens Opcenter Execution Core (Opcenter EX CR) products. REST APIs provide an easy-to-use Web API abstraction layer for executing transactions supported by the Application Server. REST APIs enable you to execute transactions more efficiently than if using WCF services.
Technical Reference: WCF Services	Describes the WCF Services provided with the Portal and the WCF Generator program used to expose the services and service methods and to generate the DLLs containing those exposed services and data contracts.

Opcenter Execution Core Intelligence

Document Name	Description
Intelligence Installation Guide	Contains information about installing Opcenter Execution Core Intelligence, and SAP® BusinessObjects™.
Intelligence Reference **Updated!**	Describes the Opcenter Execution Core Intelligence reports available for Opcenter Execution Core. It also discusses Business Objects™ universes, report guidelines, KPIs, and other related topics.

Scheduling

Document Name	Description
Scheduling User Guide	Describes how Opcenter EX CR Scheduling sends manufacturing orders and related information to Opcenter Advanced Planning and Scheduling (Opcenter APS).

Industry Solutions

Document Name	Description
Batch Processing User Guide	Contains information on batch processing and how batch processing is used to create recipes and define scales to control and track production of small chemical batches used as components of devices.
Industry Solutions User Guide	Contains information on these industry features available in the application: • Overall Equipment Effectiveness (OEE) – Provides the ability to evaluate productivity based on resource availability, resource performance in relation to capacity, and production quality. • Resource Material Queue (RMQ) – Provides the ability to track inventory quantity at an inventory location and to distribute inventory to a material queue on a production line.
Medical Device User Guide	Contains information about using functionality in the Medical Device workspace.

SPC

Document Name	Description
Inline SPC Reference Guide	Describes how to configure and test Opcenter Execution Core Inline Statistical Process Control (SPC). It also describes the various configuration options that are available to Opcenter Execution Core customers.

Manufacturing Operations Management

Document Name	Description
Teamcenter Integration Guide	Describes how to configure Opcenter Execution Core to integrate with the Siemens Teamcenter application.

Additional Documentation

Many white papers and other documents that provide in-depth information on highly specialized topics or newer features are also available on Support Center. For example, there are white papers on Containerization and documents on the Mendix-based modules.

Also, refer to Support Center for the Release Matrix and numerous JSON collections that contain examples of REST API transactions.

Siemens Support Center

Installation assistance

Contact Product Support via Support Center for assistance or to report any problems. Have your SoldTo number available. Your SoldTo number is included in the file that contains your serial number.

Website:
https://support.sw.siemens.com

Phone:
United States or Canada: +1 (800) 955-0000 or +1 (714) 952-5444

Outside the United States and Canada: Contact your local support office.

Security Information

Siemens provides products and solutions with industrial security functions that support the secure operation of plants, systems, machines and networks.

In order to protect plants, systems, machines and networks against cyber threats, it is necessary to implement – and continuously maintain – a holistic, state-of-the-art industrial security concept. Siemens' products and solutions constitute one element of such a concept.

Customers are responsible for preventing unauthorized access to their plants, systems, machines and networks. Such systems, machines and components should only be connected to an enterprise network or the internet if and to the extent such a connection is necessary and only when appropriate security measures (e.g. firewalls and/or network segmentation) are in place.

For additional information on industrial security measures that may be implemented, please visit http://www.siemens.com/industrialsecurity.

Siemens' products and solutions undergo continuous development to make them more secure. Siemens strongly recommends that product updates are applied as soon as they are available and that the latest product versions are used. Use of product versions that are no longer supported, and failure to apply the latest updates may increase customer's exposure to cyber threats.

To stay informed about product updates, subscribe to the Siemens Industrial Security RSS Feed under http://www.siemens.com/industrialsecurity.

Data Protection By Design of DI SW MOM Products and Solutions

In the course of development of DI SW MOM Products and Solutions, DI SW MOM follows the "Data protection by design" as foreseen in Article 25 of the General Data Protection Regulation ("GDPR"). This means that data protection and privacy issues are taken into account starting from the commencement of product development or solution engineering.

In general, within Siemens, the following processes are implemented:

- Data Protection by Design approach is a part of the principles actively adopted by Siemens and integrated in the secure life-cycle development of products.
- Siemens solutions adopt Threat and Risk Analysis (TRA), a Siemens-wide standardized methodology that is used for product, solution and service business during product development,

engineering or service projects. This methodology is intended to support Siemens teams in identifying typical security weaknesses and vulnerabilities, analyzing any threats that might exploit these weaknesses or vulnerabilities and evaluating any resulting risks.

Specifically for DI SW MOM products and solutions, in all data collection and processing activities that potentially involve personal data in the intended customer use case, DI SW MOM considers appropriate technical and/or organizational measures, with the goal of adequately addressing the data protection principles and safeguarding individual rights.

For Opcenter Execution Core, the following applies:

- Opcenter Execution Core has obtained the TÜV SÜD certification of security in the development process (based on IEC 62443-4-1). This standard specifies process requirements for the secure life-cycle development of products used in an Industrial Automation and Control System (IACS). The life-cycle includes:
  - the definition of security requirements;
  - secure design,
  - secure implementation (including coding guidelines), verification and validation,
  - secure defect management, patch management, and product end-of-life.
- The following personal data is processed by the Opcenter Execution Core solution:
  - The following information about the Employee Operator can be stored:
    - Username
    - Full Name
    - Login Domain
    - E-Mail Address
  - The following information regarding the Customer and Vendor can be stored:
    - Customer Name
    - Address
    - Full Name
    - Phone Number
    - Fax Number
    - Cell Phone Number
    - E-mail

Specific attention is dedicated to the processing of personal data belonging to special categories, relevant for the purpose of detecting the racial or ethnic origin, political opinions, religious or philosophical beliefs, trade union membership, genetic data, health, sexual orientation or conduct, criminal convictions and offenses or related security measures of the persons concerned. No such personal data is processed by Opcenter Execution Core.

- The personal data processed by Opcenter Execution Core is required for authentication and authorization purposes, diagnostic and log-functions. Therefore, such personal data processed by Opcenter Execution Core cannot be anonymized or pseudonymized. Such data is stored solely for these reasons, as it is necessary to identify authorized operators and record authorized actions.
- Opcenter Execution Core adopts hashing and encrypting (at rest and in transit).
- In Opcenter Execution Core, personal data can be deleted when no longer necessary for the designated purpose. In detail, the solution supports customers to define a retention period according to their needs. On the other hand, personal data may appear in the solution's data repositories as database's records, product file and log files, which are necessary for diagnostic or logging purposes. This information generated by Opcenter Execution Core has different retention periods based on their different purposes: some can be modified (or deleted) directly from the customer, others cannot be modified without the assistance of the Siemens Customer Support Service.
- Opcenter Execution Core offers Audit Trail functionalities to guarantee accuracy in the processing of personal data. An Audit Trail is a security-relevant set of records, which provide evidence of all the operations that have been carried out on a specific entity instance and detailed historical information about data changes. The customer can establish which operations on personal data are defined as Audit Trail relevant and track them in a dedicated application. In addition, hashing is used by Opcenter Execution Core to ensure integrity of personal data so that any changes can be recognized, correlated and traced, if required.
- Opcenter Execution Core Solution makes use of least privilege access control and policies and using appropriate roles and authorization concepts as individual user Read/Write permission.
- Regular review is performed to validate necessity for the purposes for which the personal data was collected and test the design against purpose limitation.

Change History

Revision	Change
Rev. 1	Original publication for Release 2510.

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About Siemens Digital Industries Software

Siemens Digital Industries Software helps organizations of all sizes digitally transform using software, hardware and services from the Siemens Xcelerator business platform. Siemens' software and the comprehensive digital twin enable companies to optimize their design, engineering and manufacturing processes to turn today's ideas into the sustainable products of the future. From chips to entire systems, from product to process, across all industries. Siemens Digital Industries Software – Accelerating transformation.

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